

Number System & HCF-LCM

Placement Aptitude Notes — TCS NQT | Accenture | Capgemini | Cognizant | Infosys | Wipro | Deloitte | EY | HCL | Tech Mahindra | LTIMindtree

Important Formulas

- Product of two numbers = HCF $\times$ LCM
- HCF of fractions = HCF(numerators) / LCM(denominators)
- LCM of fractions = LCM(numerators) / HCF(denominators)
- HCF of numbers always divides their LCM exactly
- Sum of first n natural numbers = $n(n+1)/2$
- Sum of squares of first n natural numbers = $n(n+1)(2n+1)/6$
- Sum of cubes of first n natural numbers = $[n(n+1)/2]^2$
- Number of factors of $N = a^x \cdot b^y \cdot c^z$ is $(x+1)(y+1)(z+1)$
- Sum of factors of $N = [(a^{x+1}-1)/(a-1)] \times [(b^{y+1}-1)/(b-1)] \times \dots$
- Divisibility by 2: last digit even | by 3: digit sum divisible by 3
- Divisibility by 4: last 2 digits divisible by 4 | by 5: ends in 0 or 5
- Divisibility by 6: divisible by 2 and 3 | by 8: last 3 digits divisible by 8
- Divisibility by 9: digit sum divisible by 9 | by 11: (sum of alternate digits) difference divisible by 11 (incl. 0)
- $a^n - b^n$ is divisible by $(a-b)$ always; by $(a+b)$ when n is even
- $a^n + b^n$ is divisible by $(a+b)$ when n is odd

Properties & Shortcuts

- To find HCF: use Euclid's Division Algorithm — repeatedly divide larger by smaller till remainder = 0.
- To find LCM of numbers with common factors: LCM = product of all prime factors taken with highest powers.
- HCF of two consecutive integers is always 1 (co-prime).
- HCF and LCM of fractions: convert all to simplest form first.
- Number of trailing zeros in $n!$ = $\text{floor}(n/5) + \text{floor}(n/25) + \text{floor}(n/125) + \dots$
- To find unit digit of a^n : find cyclicity of a (cycle length 1, 2 or 4) and use $n \bmod \text{cycle}$.
- Cyclicity table: $2 \rightarrow 4, 3 \rightarrow 4, 4 \rightarrow 2, 5 \rightarrow 1, 6 \rightarrow 1, 7 \rightarrow 4, 8 \rightarrow 4, 9 \rightarrow 2, 0/1 \rightarrow 1$.
- Prime numbers below 100: 25 primes (2,3,5,7,...,97) — memorize for quick factorization.
- Every prime number >3 can be written as $6k \pm 1$.
- Co-prime (relatively prime) numbers have HCF = 1 (need not be prime themselves).
- To check if N is prime: test divisibility only up to $\sqrt{N}$.
- Remainder shortcut: $(a \times b) \bmod m = [(a \bmod m) \times (b \bmod m)] \bmod m$.

Previous-Year Frequently Asked Concepts

- Finding HCF/LCM of 2-4 numbers directly or via word problems (bells ringing together, tanks filling, traffic lights).
- Least/Greatest number leaving a given remainder when divided by given numbers (LCM $\pm$ remainder type).
- Number of factors, sum of factors, and number of even/odd factors of a given integer.
- Unit digit of large powers (a^n type) and unit digit of expressions/products.
- Number of trailing zeros in factorials ($n!$).
- Divisibility rule based questions (checking divisibility of large numbers by 3, 9, 11, etc.).
- Remainder theorem based questions (finding remainder of large powers divided by small numbers).
- Co-prime number counting (numbers less than N and co-prime to N — Euler's totient concept).
- Sum/product-based word problems involving HCF and LCM together (product = HCF $\times$ LCM).

Solved Examples

Ex 1: Find HCF and LCM of 36 and 60.

Solution: $36 = 2^2 \times 3^2$, $60 = 2^2 \times 3 \times 5$. $\text{HCF} = 2^2 \times 3 = 12$. $\text{LCM} = 2^2 \times 3^2 \times 5 = 180$. Check: $12 \times 180 = 2160 = 36 \times 60$ ✓

Ex 2: Three bells ring at intervals of 12, 15 and 18 minutes. If they ring together at 9:00 AM, when will they ring together again?

Solution: $\text{LCM}(12, 15, 18)$: $12 = 2^2 \times 3$, $15 = 3 \times 5$, $18 = 2 \times 3^2$. $\text{LCM} = 2^2 \times 3^2 \times 5 = 180$ minutes = 3 hours. Next ring together at 12:00 PM.

Ex 3: Find the least number which when divided by 6, 8 and 12 leaves remainder 3 in each case.

Solution: $\text{LCM}(6, 8, 12) = 24$. Required number = $24 + 3 = 27$.

Ex 4: Find the unit digit of 7^{95} .

Solution: Cyclicity of 7 is 4: (7, 9, 3, 1). $95 \bmod 4 = 3 \rightarrow$ 3rd value in cycle = 3. Unit digit = 3.

Ex 5: Find the number of trailing zeros in $100!$.

Solution: $\text{floor}(100/5) + \text{floor}(100/25) + \text{floor}(100/125) = 20 + 4 + 0 = 24$ trailing zeros.

Practice Questions (1–30)

- Q1.** Find the HCF of 84 and 126.
- Q2.** Find the LCM of 15, 20 and 25.
- Q3.** The HCF and LCM of two numbers are 6 and 210 respectively. If one number is 30, find the other.
- Q4.** Find the greatest number that divides 285 and 1249 leaving remainders 9 and 7 respectively.
- Q5.** Find the least number which when divided by 5, 6, 7 and 8 leaves remainder 3 in each case.
- Q6.** Three bells toll at intervals of 9, 12 and 15 minutes respectively. If they toll together at 8:00 AM, at what time will they next toll together?
- Q7.** Find the smallest number which when increased by 5 is divisible by 24, 32 and 36.
- Q8.** Find the number of factors of 360.
- Q9.** Find the sum of all factors of 60.
- Q10.** Find the unit digit of 3^{57} .
- Q11.** Find the unit digit of $24^{36} \times 17^{47}$.
- Q12.** Find the number of trailing zeros in $50!$.
- Q13.** Find the number of trailing zeros in $125!$.
- Q14.** How many numbers between 1 and 100 are divisible by both 3 and 5?
- Q15.** Two numbers are in the ratio 4:5 and their HCF is 6. Find their LCM.
- Q16.** Find the HCF of $\frac{2}{3}$, $\frac{4}{9}$ and $\frac{6}{27}$.
- Q17.** Find the LCM of $\frac{3}{4}$, $\frac{5}{6}$ and $\frac{7}{12}$.
- Q18.** What is the largest 4-digit number exactly divisible by 88?
- Q19.** What is the smallest 5-digit number exactly divisible by 476?
- Q20.** Find the remainder when 2^{100} is divided by 7.
- Q21.** Find the remainder when 7^{105} is divided by 5.
- Q22.** How many numbers from 1 to 200 are co-prime to 200?
- Q23.** A number when divided by 296 leaves remainder 75. What remainder is obtained when the same number is divided by 37?
- Q24.** Find the sum of first 40 natural numbers.
- Q25.** Find the sum of squares of first 15 natural numbers.
- Q26.** If $N = 2^3 \times 3^2 \times 5$, find the number of even factors of N.
- Q27.** Find the HCF of 96, 144 and 192.
- Q28.** The product of two numbers is 1320 and their HCF is 6. Find their LCM.
- Q29.** Find the least number which is a perfect square and is divisible by 16, 18 and 45.
- Q30.** Check whether 1001 is divisible by 11. If yes, find the quotient.

Answer Key

Q.No	Answer
Q1	42
Q2	300
Q3	42
Q4	138
Q5	843
Q6	11:00 AM
Q7	283
Q8	24
Q9	168
Q10	3
Q11	8
Q12	12
Q13	31
Q14	6
Q15	120

Q.No	Answer
Q16	$2/27$
Q17	$105/2$
Q18	9944
Q19	10472
Q20	2
Q21	2
Q22	80
Q23	1
Q24	820
Q25	1240
Q26	18
Q27	48
Q28	220
Q29	3600
Q30	Yes, quotient = 91